PREGNANCY ABDOMINAL SUPPORT OPTIMIZER

Problem Statement, Existing-Product Gap & User Journey

Project framing
A biomedical engineering prototype exploring pressure-sensing and personalized fit feedback in a wearable pregnancy support belt.

1. Background and problem context

Pregnancy can change abdominal load, posture, balance and the demands placed on the lower back and pelvis. Some people use maternity support belts to make everyday activities—such as standing, walking or completing routine tasks—more comfortable. Commercial belts commonly rely on fabric panels, elastic bands and hook-and-loop straps to provide adjustable mechanical support. For example, products such as the Frido Maternity Belt advertise belly and lower-back support, an adjustable fit as the bump grows, and breathable fabric [1]. Mowell also markets an adjustable abdominal support belt [2].

However, adjustment is largely based on the wearer's feel and judgement. The wearer may not know whether the belt is sitting as intended, whether support is distributed consistently across the belt's contact regions, or whether a change in posture or activity has changed that distribution. A belt can be adjustable without objectively measuring its fit. This creates an engineering opportunity to investigate whether simple pressure sensing and understandable feedback can help users assess belt fit more consistently.

Core problem: Existing pregnancy support belts can offer adjustable mechanical support, but typical designs do not provide direct, real-time measurement of pressure distribution across multiple belt regions. Users therefore have limited objective feedback to guide fit adjustments as their body shape, posture and daily activities change.

2. Problem statement

To design and prototype a wearable pregnancy abdominal support system that measures relative pressure at multiple locations along the belt and provides simple fit-related feedback, with the aim of exploring a more informed and personalized adjustment process. The prototype will use an adjustable fabric belt, a distributed sensor layer and detachable electronics. It will be evaluated initially using a bench setup, instrumented form/mannequin or other non-pregnant test fixture, rather than testing on pregnant participants.

3. Why this problem matters

A one-size-fits-many mechanical design may not feel the same for every wearer, and the body changes throughout pregnancy. A pressure-sensing prototype could help make fit information more visible than relying on strap tension or intuition alone. This is relevant to biomedical engineering because it combines wearable design, sensing, human factors, and the interpretation of biomechanical measurements. It also creates a testable engineering question: can a low-cost array of sensors detect repeatable changes in pressure distribution when the belt is repositioned or adjusted?

Important distinction: The project does not assume that a particular pressure pattern is medically optimal. It investigates sensing and feedback as a prototype feature; safe pressure

limits, clinical recommendations and pregnancy-specific efficacy would require qualified clinical input and appropriate validation.

4. Existing products and the design gap

The following comparison is based on publicly available product descriptions and should be read as a high-level market scan, not a complete survey of every product. The listed limitations are design opportunities to investigate, not claims that every product is uncomfortable or ineffective.

Product / category	Features described publicly	Potential gap relevant to this project
Frido Maternity Belt [1]	Two-part abdominal and lower-back support; hook-and-loop adjustment; breathable knit fabric.	The public description emphasizes mechanical adjustment and materials; it does not describe multi-point pressure sensing or measured pressure-distribution feedback.
Mowell Maternity Support Belt [2]	Elastic fabric and adjustable Velcro strap; marketed for abdominal support and comfort.	The public listing does not describe sensor-based fit measurement or a feedback loop that responds to measured pressure across zones.
Conventional maternity belts (general category) [3, 4]	Usually provide support through garment geometry, elastic tension and manual adjustment.	Fit decisions are typically subjective; adjustment may need to be repeated as the wearer changes position or activity. This should be tested rather than assumed for every design.

Evidence and interpretation

Published reviews caution that the evidence for maternity support garments is limited and varies by product, study design and outcome. A 2019 systematic review found that only a small number of studies had evaluated garments as a standalone intervention and called for further objective research [3]. A 2025 systematic review and meta-analysis of pelvic belts reported modest pain reduction and minimal improvement in disability, with low to very low certainty of evidence [4]. These findings do not mean that belts never help; they mean that benefit should not be assumed for every user or design.

Specific opportunity for the proposed prototype

The intended differentiator is not simply that the belt is 'smart'. Sensorized wearables already exist in adjacent areas, so the project’s more defensible focus is a low-cost, multi-zone pressure measurement and fit-feedback workflow for a pregnancy support belt. The prototype can explore whether sensor readings are repeatable, whether changes in fit are detectable, and whether feedback is understandable to a user. It should not claim automatic clinical optimization unless an appropriate validated method and evidence are developed.

5. Proposed design response

Concept: An adjustable fabric belt with an embedded layer of eight FSR 402 force-sensing points, read through two ADS1115 analogue-to-digital converter modules and an ESP32-S3-DevKitC-1-N8 controller. LEDs provide the initial visual feedback; vibration feedback may be explored as an optional feature. The first prototype is planned as a USB-powered bench system, with battery integration considered later.

Potential prototype functions

• Measure relative sensor readings at multiple locations around the belt.

• Show a simple zone-level indication when readings differ from a chosen, experimentally established reference pattern.

• Allow the user to loosen, reposition or re-seat the belt and check whether the readings change.

• Record test data to evaluate repeatability across known fixture positions and strap adjustments.

The sensor array should initially be treated as a relative measurement system, not a calibrated medical pressure monitor. FSR response can depend on sensor placement, curvature, fabric layers, loading history and calibration; these effects should be documented during testing.

6. User journey

The journey below describes the intended interaction for a future user-facing prototype. In early development, the same sequence should be tested on a mannequin or non-pregnant form rather than on pregnant participants.

1. Need identified

The user experiences a need for additional abdominal or lower-back support during an everyday activity and considers a support belt.

2. Select and inspect

The user selects an appropriate belt size and checks fabric condition, closures and the device's instructions. Clinical concerns are referred to a qualified healthcare professional.

3. Put on the belt

The user positions the belt according to the instructions and fastens it comfortably without deliberately tightening it to chase a sensor reading.

4. Take a baseline reading

The sensor layer collects readings from the eight sensing points. The system displays a simple zone map or indicator; the reading is treated as fit feedback, not a diagnosis.

5. Review feedback

If the prototype indicates a large difference from its experimentally defined reference pattern, the user is prompted to check placement and comfort—not to increase compression automatically.

6. Adjust gently

The user may reposition the belt or make a small adjustment, following the product instructions, and then take another reading. The user can stop if the belt causes discomfort.

7. Check during activity

For bench testing, readings are collected under defined fixture positions or simulated movement. Any future real-world use would need safety review and appropriate validation before it could be recommended.

8. Finish and clean

The user removes the belt, disconnects the electronics if needed, and follows fabric-care instructions. The prototype's electronics and sensors are checked for damage before the next test.

7. User needs and design requirements

User need	Design implication	How to evaluate in prototype
Comfort and adjustability	Soft, adjustable belt; no instruction to tighten in response to a high reading.	Fit and placement checks on a non-human fixture; record pressure changes during adjustment.
Understandable feedback	Simple LED or zone indication rather than raw numbers alone.	Ask test observers to interpret the display; record errors or ambiguity.
Repeatable measurements	Sensor mounting and wiring should resist shifting during controlled tests.	Repeat measurements under the same fixture setup and compare variability.

User need	Design implication	How to evaluate in prototype
Practical wearability	Detachable electronics, manageable wiring and a washable fabric section where feasible.	Inspect setup time, cable interference and ease of assembly/disassembly.

8. Scope, assumptions and validation plan

In scope

• Build an adjustable belt prototype with eight pressure-sensing points, two ADS1115 modules and an ESP32-S3 controller.

• Create basic LED-based feedback and a documented method for collecting sensor readings.

• Characterize sensor repeatability and response to controlled placement and strap-tension changes on a fixture.

• Document design limitations, usability considerations and next steps.

Out of scope for the initial prototype

• Diagnosing pregnancy-related conditions or prescribing a pressure distribution.

• Claiming to prevent injury, improve fetal outcomes, correct posture, or treat pain.

• Testing the prototype on pregnant participants without ethics approval, appropriate clinical supervision and a formal safety protocol.

• Automatic tightening or compression adjustment based only on sensor readings.

Proposed validation sequence

Stage 1 — Bench verification: check wiring, ADC readings, LED states and basic fault handling.
Stage 2 — Sensor characterization: compare readings under repeatable known loading conditions; document drift, hysteresis and differences between sensors.
Stage 3 — Fit-pattern tests: use a mannequin or curved fixture to compare sensor patterns across predefined belt positions and adjustment settings.
Stage 4 — Feedback review: assess whether the LED indications are understandable and whether the suggested action is safe and non-prescriptive.
Stage 5 — Future work: consult relevant clinicians and human-factors specialists before any study involving pregnant participants.

9. Expected outcome

The expected outcome is a proof-of-concept system that can capture relative pressure readings at multiple belt locations and present simple fit-related feedback. Success at this stage means demonstrating reliable sensing under controlled conditions and a clear user interaction—not demonstrating clinical effectiveness. The work can establish a technical foundation for later studies on calibration, comfort, human factors and clinically appropriate use.

10. References and market sources

[1] Frido, “Frido Maternity Belt” — product description, adjustment features and manufacturer safety note.
https://myfrido.com/products/frido-maternity-belt

[2] Tata 1mg, “Mowell Maternity Support Abdominal Belt” — public product listing.
https://www.1mg.com/otc/mowell-maternity-support-abdominal-belt-l-xl-otc956447

[3] Quintero Rodriguez et al. (2019), “The Effect of Maternity Support Garments on Alleviation of Pains and Discomforts during Pregnancy: A Systematic Review,” Journal of Pregnancy.
https://pubmed.ncbi.nlm.nih.gov/31467715/

[4] “The effect of pelvic belts to manage low back and pelvic pain during pregnancy: a systematic review and meta-analysis” (2025), PubMed.
https://pubmed.ncbi.nlm.nih.gov/40716213/

[5] Ho et al. (2009), “Effectiveness of maternity support belts in reducing low back pain during pregnancy: a review,” Journal of Clinical Nursing.
https://pubmed.ncbi.nlm.nih.gov/19490291/

Use note: This document is a project-planning draft, not clinical guidance. Product features and availability can change; market observations are based on the linked public pages reviewed for this draft. Any future human testing, pregnancy-specific recommendations or claims of health benefit require appropriate professional oversight and validation.